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BOOK REVIEW

A taxonomy of pseudosciences

Michael D. Gordin: On the fringe: where science meets pseudoscience. Oxford: Oxford University Press, 2021, 122 pp, $18.95 HB

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There are many examples in the philosophy of science of non-philosophical approaches to the nature of science yielding very suggestive results. Among these, the 'historical turn' initiated in the 1960s stands out. Historians of science and historically minded philosophers such as Thomas Kuhn and Imre Lakatos revolutionized the way we think about rationality or scientific change. But if there is a quintessential philosophical problem for those interested in how science works, it is the so-called demarcation problem, or the attempt to pinpoint the specific characteristics of science that distinguish it from other forms of nonscientific knowledge and, particularly, pseudoscience.

In the tradition of the historical turn, Michael D. Gordin's *On the fringe* constitutes an attempt to provide a historically informed approach to this quintessential philosophical problem. Gordin, a professor of history at Princeton University, shows how most attempts at demarcation so far have not taken the historical diversity of 'fringe doctrines,' as he calls them, seriously enough. Most of them have been conceived inductively based on one or a few representative instances of pseudoscience, thereby leading to highly unsatisfactory criteria. Given the astounding historical diversity of both science and pseudoscience, a much more productive approach is to devise what Gordin calls 'local demarcation criteria,' that is, "characterizations that encompass groupings of fringe doctrines without claiming to provide a be-all, end-all solution to the demarcation problem" (12). In this sense, the main aim of *On the fringe* is to provide a historically accurate taxonomy of all those theories whose scientific status is or has been under dispute.

The result of such a taxonomy is four broad classes of fringe doctrines: (1) vestigial sciences, (2) hyperpoliticized sciences, (3) counterestablishment sciences, and (4) theories that posit extraordinary powers of the mind. Although Gordin acknowledges that such categorization might be limited in some respects, at least

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it can provide a starting point for analyzing the tremendous historical diversity of pseudosciences.

After an introductory chapter contextualizing the demarcation problem and showing why, despite its popularity, the most famous demarcation criterion so far, Popper's falsificationism, fails, Gordin devotes a chapter to each of these four families of fringe doctrines. Thus, Chapter 2 deals with 'vestigial sciences.' This family of doctrines is characterized not by *what* they claim, but by *when* they do it. Science is a dynamic enterprise characterized by a constant flux of ideas: old theories are abandoned, and new ones come to take their place. This means that if a group of people continues to stick to ideas that have already been discarded by the scientific community, then it is highly likely that they will be called pseudoscientists. As examples of vestigial sciences, Gordin describes the cases of astrology and alchemy, providing very rich—yet concise—accounts of how these doctrines transitioned from well-respected endeavors to outdated doctrines.

Chapter 3 deals with Gordin's second family of fringe doctrines, namely, hyperpoliticized sciences. Even though all science may be somewhat politicized, when this politicization is taken to the extreme it leads to doctrines that can become, in Gordin's words, 'arms of a particular political ideology,' and whose epistemic status is highly controversial. Gordin illustrates this category of doctrines with three examples: Aryan Physics, Lysenkoism, and Eugenics. When it comes to Aryan Physics, he describes the role of Philipp Lenard and Johannes Stark as scourges of the so-called Jewish Physics, which they decried for being 'too mathematical'. As for Lysenkoism, the author narrates the story behind the rise and fall of Trofim Lysenko's doctrine, a sort of neo-Lamarckian agrobiology that contributed to the proscription of classical genetics up until 1965. Lastly, with the case of eugenics Gordin shows that, even if hyperpoliticized sciences might be automatically associated with authoritarian regimes, liberal democracies can also sometimes nurture these kinds of doctrines.

Chapter 4 presents a third group of pseudosciences whose main characteristic is a constant fight with the scientific establishment. Counterestablishment pseudoscientists consider that the mainstream scientific consensus is wrong and mostly driven by non-epistemic considerations. To have a chance in their fight with the establishment, counterestablishment scholars try to replicate as faithfully as possible the institutional structures of mainstream science: they create conferences and specialized journals, and, if possible, set up degree programs and research institutes. Because of their emphasis on institutional articulation, Gordin points out that counterestablishment sciences are more visible to the broad public than other fringe doctrines. In his catalog, Gordin describes six counterestablishment pseudosciences that have had uneven success in their attempt to create parallel scientific institutions: phrenology, creationism, cryptozoology, cosmic catastrophism, ufology, and flat earth theory.

The fourth and last family of pseudosciences is discussed in Chapter 5 and groups all those doctrines or disciplines that attribute extraordinary powers to the mind. These include Mesmerism, spiritualism, and the more recent parapsychology. Especially interesting is the history of the latter, which acquired a relatively high reputation thanks to the work of Joseph Banks Rhine, a botanist turned parapsychologist from Duke University who conducted research on extrasensory perception.

Gordin shows that, despite its fringe nature, parapsychology had a tremendous positive impact on mainstream psychology, as attempts at debunking parapsychological claims by skeptics launched a methodological arms race that "pushed psychology to ever greater sophistication in both the laboratory and in data analysis" (60).

The last two chapters of *On the fringe* are devoted to discussing a series of questions prompted by the previous historical analysis of pseudoscience. The title of Chapter 5 is telling of one of Gordin's main conclusions: 'Controversy is inevitable.' The very nature of science necessarily leads to the emergence of new forms of pseudoscience. However, there is a continuum between ideas that, although rejected, are considered legitimate science and those that constitute the seeds of a future pseudoscience. Gordin explains this continuum with three cases of rejected scientific ideas whose scientific status varied with time: polywater, water memory, and cold fusion.

Finally, Chapter 6 deals with what Gordin calls 'the Russian questions' after the title of two classic Russian novels: *Who is to blame?* and *What is to be done?*. We know much more about nature than ever before, yet fringe doctrines are still around. This might call for not only responsibilities but also measures to counteract them. As for the first, Gordin points out that even if our primary, intuitive move is to blame the pseudoscientists themselves, this is likely to lead to a dead-end. A more useful approach is to provide 'local' criteria for responsibilities. To illustrate this, Gordin analyzes the denialist movements, from the corporation-driven instances of doubt-mongering to the more grass-roots efforts of the anti-vaxx activists. To the second Russian question, 'What is to be done?,' Gordin argues that neither debunking nor improving science education are satisfactory solutions. Tightening or relaxing the standards for scientific publication is also discouraged, as they can either stifle scientific innovation or lead to the rise of even more fringe doctrines. Instead, more useful is to get rid of the 'publish or perish' culture, "an imposition which taxes the whole system's ability to evaluate the credibility of findings and at the same time provides incentives for fraudulent or slipshod research" (100).

Gordin ends *On the fringe* by emphasizing one idea that permeates all chapters, namely, that pseudoscience is an unavoidable consequence of science. The only way of definitely getting rid of pseudoscience is by eliminating science, something that no one in their right mind would accept. Our task should be, therefore, to try to understand the diversity of pseudosciences to set apart those that might pose threats to the public interest from those unthreatening oddities. Pseudoscience is the shadow of science, but as Gordin puts it, 'not all shadows hide monsters,' so not all fringe doctrines deserve to be treated harshly.

Gordin's book should be mandatory reading for all those interested in the nature of science and pseudoscience. Unlike most philosophical discussions of the demarcation criteria, *On the fringe* provides an excellent exposition of a wide diversity of pseudoscientific doctrines, something which certainly can help to devise more useful demarcation criteria. However, as Gordin himself acknowledges, his approach is limited in that the explicit absence of methodological criteria for demarcating science from pseudoscience ends up generating a categorization of fringe doctrines whose classes significantly overlap and which is built around highly heterogeneous criteria. Furthermore, in some parts of the book, the excessive attention drawn to the historical description of doctrines might prevent the reader from seeing the forest for

the trees. In this sense, Gordin's book can best be approached as a first sketch of a very useful and promising way of studying pseudoscience rather than as a definitive account of it.

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